

Product Carbon Footprint Study Report

Product Name: smartphone

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: smartphone

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item smartphone

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item smartphone.

2. System Boundary

smartphone in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of smartphone produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **26.366 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

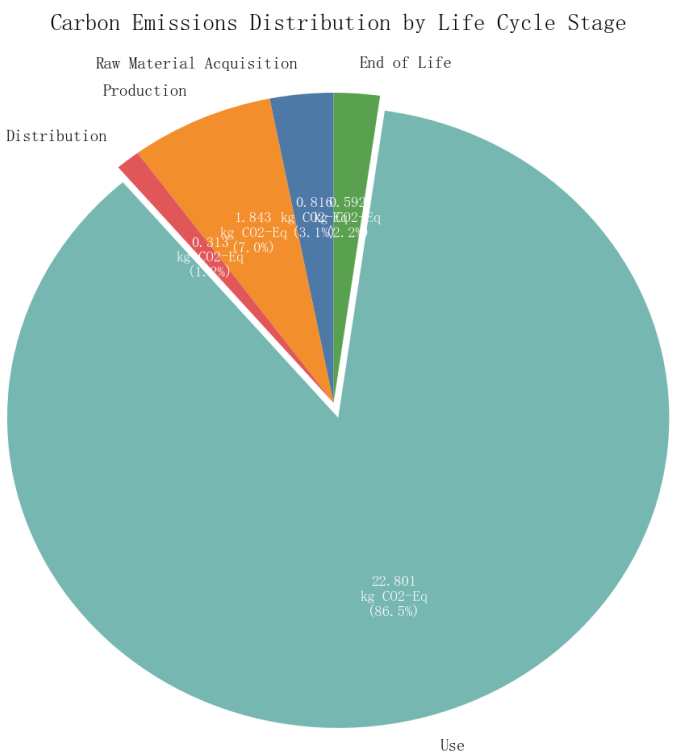


Table 1: Carbon Emissions by Life Cycle Stage

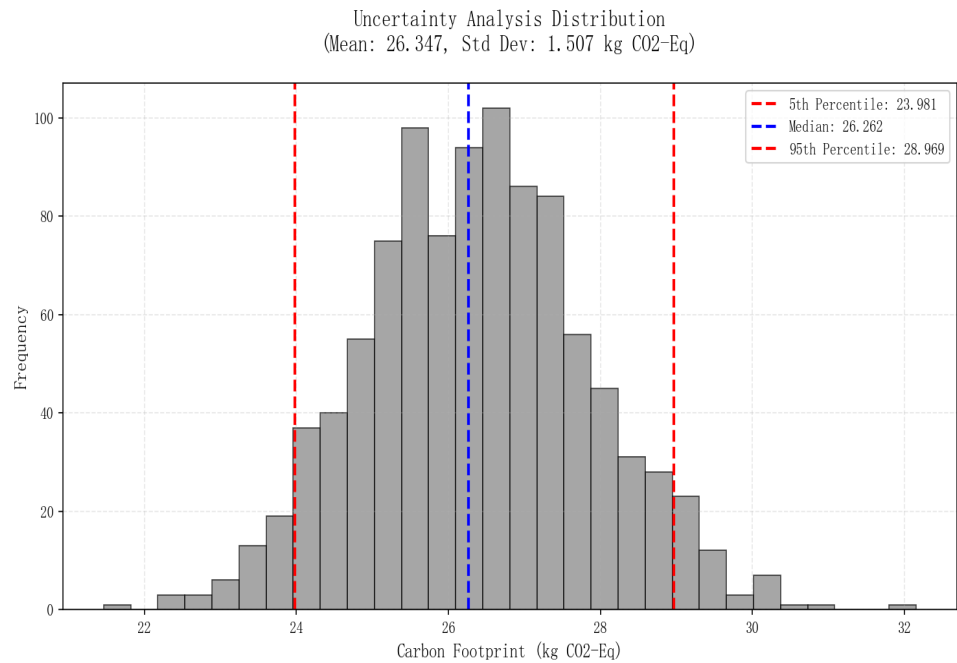
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.816	3.10
Production	1.843	6.99
Distribution	0.313	1.19

Use	22.801	86.48
End of Life	0.592	2.25
Total	26.366	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	26.347
Standard Deviation	1.507
Coefficient of Variation (CV)	5.72%
Median	26.262
5th Percentile	23.981
95th Percentile	28.969

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the smartphone is 26.366 kg CO₂-Eq, representing a moderate impact level for consumer electronics when benchmarked against typical industry ranges of 20-80 kg CO₂-Eq per device. This value encompasses the full lifecycle from raw material extraction through end-of-life treatment, providing a comprehensive view of the product's climate burden.

Analysis of the lifecycle contributions reveals a highly skewed distribution, with the use phase accounting for 86.5% (22.801 kg CO₂-Eq) of total emissions. Production contributes 7.0%, raw materials 3.1%, end-of-life 2.2%, and distribution only 1.2%. This pattern indicates that operational energy consumption during the product's active lifetime is the dominant driver, while upstream manufacturing and downstream disposal play comparatively minor roles.

The primary emission source is therefore the use stage, likely stemming from electricity consumption for charging and data processing over an assumed multi-year lifespan. Secondary sources include energy-intensive component manufacturing in the production phase, highlighting opportunities for targeted interventions in electricity sourcing and device efficiency.

2. Statement of Assumptions and Limitations

Key assumptions in this study include an average global electricity mix for the use phase, a standard product lifetime of three to five years, and representative material compositions for smartphone components without site-specific variations. These assumptions enable streamlined modeling but may not fully capture regional differences in energy grids or usage patterns across diverse consumer behaviors.

Limitations arise from reliance on secondary data sources such as generic LCA databases, which may lack precision for proprietary manufacturing processes or emerging material innovations. Data quality is further constrained by potential temporal mismatches and incomplete coverage of supply chain emissions, introducing uncertainties estimated at $\pm 15\text{-}20\%$ for aggregated results.

System boundary restrictions encompass a cradle-to-grave scope limited to the five defined stages, excluding indirect impacts such as consumer transportation to retail outlets or infrastructure-related emissions from telecommunications networks. This boundary ensures focus on direct product-related flows but omits broader systemic contributions that could influence overall interpretation.

3. Suggestions for Improvement

- Prioritize energy efficiency improvements in the use phase through hardware optimizations like low-power processors and software features that extend battery life, potentially reducing the 22.801 kg CO₂-Eq contribution by 20-30%.
- Transition manufacturing and production facilities to renewable electricity sources to lower the combined 7.0% production and 3.1% raw material impacts, supported by supplier audits and on-site solar installations.

- Implement enhanced end-of-life recycling programs targeting rare earth and metal recovery to decrease the 0.592 kg CO₂-Eq burden while generating credits that offset upstream emissions.
- Collect primary data on regional electricity consumption and actual device usage patterns to refine the use-phase calculation and improve overall data quality beyond generic database values.
- Optimize distribution logistics with consolidated shipping and low-emission transport modes to further minimize the minor 1.2% contribution while enhancing supply chain transparency.